

Peer Review Report

Manuscript Title: *AI-Assisted Programming in Software Engineering: A Systematic Literature Review and Socio-Technical Perspective*

Journal: *Empirical Software Engineering*

Recommendation: Major Revisions

Overall Manuscript Rating: 62/100

General Assessment

This manuscript addresses an important, timely, and rapidly evolving topic: the empirical consequences of AI-assisted programming for software development processes and outcomes. Its central ambition is valuable. Rather than treating AI-assisted programming merely as a productivity tool, the manuscript attempts to synthesize evidence across technical, human, organizational, cognitive, and economic dimensions and to formalize these tensions through a “socio-technical equilibrium” framework. This is a promising contribution, especially for a journal such as *Empirical Software Engineering*, where the readership expects rigorous empirical synthesis with practical relevance to software engineering research and practice.

However, the manuscript is not yet ready for publication in its current form. The most serious concerns are methodological rather than topical. The systematic review process is insufficiently transparent and contains internal inconsistencies, especially regarding the use of AI tools in screening, extraction, thematic exploration, and synthesis. The quality-assessment framework is under-specified, the distinction between CORE, SUPPORTING, PERIPHERAL, and REMOVE CANDIDATE studies is not operationalized with sufficient precision, and several claims are stronger than the underlying evidence can support. The manuscript also relies heavily on descriptive percentages drawn from heterogeneous studies without always clarifying denominators, study designs, uncertainty, or comparability. The discussion is intellectually ambitious but sometimes rhetorically overextended, and the figures/tables need refinement for clarity, reproducibility, and evidentiary value.

The manuscript has substantial potential, but it requires a rigorous methodological revision, sharper claim calibration, a clearer audit trail, and significant tightening of presentation before it can be reconsidered.

Fit with the Journal

The manuscript is broadly aligned with *Empirical Software Engineering* because it addresses empirical evidence on software development tools, processes, quality, human-AI interaction, productivity, and industrial adoption. The topic is clearly relevant to the journal's readership, especially given the growing use of AI coding assistants in professional software engineering.

The fit would be substantially stronger if the manuscript more clearly met the reporting expectations of empirical software engineering and evidence-based software engineering. At present, the paper presents

itself as a systematic literature review following PRISMA 2020 and Kitchenham-style guidelines, but the methodological reporting does not yet provide enough detail for independent replication. For this journal, the authors should strengthen the review protocol, search transparency, screening documentation, quality-assessment rubric, data-extraction procedures, and supplementary materials. The socio-technical framework is appropriate for the journal, but it must be grounded more explicitly in evidence-synthesis logic rather than presented mainly as an interpretive narrative emerging from the corpus.

Main Strengths

The manuscript's strongest contribution is its attempt to move beyond the simplistic question of whether AI coding assistants "increase productivity." The distinction between generation efficiency and verification overhead is conceptually useful and reflects an important empirical tension in the literature. The manuscript is also right to emphasize that productivity, code quality, security, maintainability, trust, cognitive load, and organizational governance cannot be interpreted independently.

The size of the reported corpus is another strength. A review claiming 312 empirical studies, plus 67 related or supporting sources, has the potential to provide a broad and useful map of a rapidly expanding research area. The manuscript also does well in identifying underdeveloped areas of the literature, especially economic impact, organizational impact, and cognitive load. The explicit recognition that economic evidence is thin is commendable and prevents the paper from overstating ROI claims.

The threats-to-validity section is unusually candid in some respects. The manuscript acknowledges the high proportion of preprints, the single-author nature of the review, the absence of inter-rater reliability, the use of AI-assisted translation, and the temporal fragility of findings in a fast-moving field. These are important disclosures. However, as explained below, the same section also reveals inconsistencies with the earlier methodology section that must be corrected.

Major Comments to the Authors

1. The systematic review protocol is insufficiently reproducible.

The manuscript states in Section 2 that the review followed PRISMA 2020 and evidence-based software engineering guidelines. However, the search process is not described with enough precision to allow replication. The manuscript reports that Branch A used SciSpace, including Deep Search, Full-Text Search, and programmatically queried Google Scholar, while Branch B used Scopus, ProQuest, and arXiv. Yet the main manuscript does not provide the exact search strings, Boolean operators, field restrictions, language restrictions, export formats, full search dates for all sources, or database-specific filters.

This matters because the review's central empirical authority depends on the corpus-construction process. The use of SciSpace as the primary pathway is especially consequential. AI-assisted discovery tools can be useful, but their indexing, ranking, retrieval, and scoring procedures are not fully transparent. The manuscript states that SciSpace applied title/abstract and full-text scoring thresholds of 4.0 and 4.5, respectively, but it does not explain how those scores were generated, what scale they used, whether the author calibrated them, or how many manually overridden decisions occurred.

The authors should add a detailed search appendix in the manuscript or supplementary material and summarize it more concretely in Section 2. At minimum, this should include exact queries, database fields searched, filters, search dates, number of records per query, deduplication procedure, title/abstract screening criteria, full-text screening criteria, and a transparent explanation of how SciSpace scores were used. The manuscript should also clarify whether SciSpace's Google Scholar retrieval is reproducible and whether the authors independently validated records retrieved through that route.

2. There is a serious contradiction regarding the use of AI tools in the review process.

Section 2.4 states that AI tools were not used to perform synthesis, data extraction, or interpretation of findings. This is a strong and reassuring claim. However, Section 5 states that NotebookLM was used for corpus-wide thematic exploration and descriptive characterization, ChatGPT was used for single-paper deep extraction and scaffolding artifacts, and Elicit was used for cross-validation of selected quantitative extractions. The same section also reports observed AI-tool errors in extraction, including conflating cited findings with a paper's own findings and extracting bracketed reference numbers as quantitative findings.

This is the most important internal inconsistency in the manuscript. If AI tools were used for "single-paper deep extraction," "corpus-wide thematic exploration," and "selected quantitative extractions," then Section 2.4 cannot state without qualification that AI tools were not used for data extraction or synthesis. The issue is not necessarily that AI tools were used; the issue is that their role is described inconsistently.

The authors should revise Section 2.4 and Section 5 into a single transparent account of AI use across the review workflow. They should specify which tools were used, for which tasks, at which stage, using which safeguards, and what was manually verified. If AI-assisted outputs informed thematic coding, extraction tables, classification, or synthesis, this must be stated clearly. The authors should also provide representative prompts, validation procedures, error rates, and examples of corrected misattributions. Without this correction, the methodological integrity of the review is weakened.

3. Corpus construction and record counts need clearer presentation.

The manuscript contains many record-count statements, and most appear arithmetically plausible, but their presentation is confusing. Table 1 reports final operational categories: Corpus = 312, Related Work or Support = 67, Duplicated = 21, No Full Text = 22, and Excluded = 5. These sum to 427. Section 2.7, however, explains that the structured search produced 418 records, which were then distributed as 308 Corpus, 62 Related Work or Support, 21 Duplicated, 22 No Full Text, and 5 Excluded. A supplementary discovery procedure then added 4 Corpus records and 5 Related Work records, producing final totals of 312 and 67.

The arithmetic can be reconciled, but the current organization makes the process harder to follow than necessary. Table 1 appears before the full PRISMA explanation, which makes it unclear whether the categories refer to the structured-search dataset, the final dataset after supplementary discovery, or both. The authors should revise Table 1 or add a second table that explicitly distinguishes: structured-search records, supplementary records, final taxonomy, and final analytical corpus. This would also clarify whether "No Full Text" and "Excluded" records are part of the retained dataset, part of the audit trail, or part of the final operational archive.

The manuscript should also clarify the 2021–2026 window. The abstract and methodology state that studies from 2021 to 2026 were targeted, but the final corpus spans 2022 to 2026. If no 2021 studies met the

criteria, this should be explicitly stated. Similarly, because 38.8% of the corpus consists of preprints, the manuscript should avoid language suggesting that all corpus items are “published studies” in the conventional peer-reviewed sense.

4. The quality-assessment framework is under-specified and potentially subjective.

Section 2.8 states that studies were classified as A1, A2, or B based on empirical rigour, methodological strength, venue quality, reporting completeness, and relevance. However, the manuscript does not provide a scoring rubric, decision rules, examples, thresholds, or item-level coding criteria. The definitions are too broad. For example, A1 includes controlled experiments, large-scale empirical studies, and causal studies; A2 includes user studies, case studies, and structured evaluations; B includes perception studies, qualitative analyses, and model-based approaches. These categories conflate design type, sample size, causal leverage, publication venue, and perceived evidentiary strength.

This matters because the paper repeatedly relies on the A1/A2/B classification to claim that the evidence base is methodologically strong and that some dimensions are more rigorous than others. It also claims that the share of A1 studies increased from 16.7% in 2022 to 54.7% in 2026, implying methodological maturation. That conclusion may be valid, but it cannot be evaluated without knowing how A1 was assigned and whether the criteria were applied consistently.

The CORE/SUPPORTING/PERIPHERAL/REMOVE CANDIDATE segmentation raises a similar concern. The manuscript says this segmentation governs interpretive weight, but it does not sufficiently define the criteria for each category. The inclusion of six “REMOVE CANDIDATE” studies inside the Corpus is especially problematic. If these papers are candidates for removal, the authors should either exclude them from the corpus or explain why they remain in the descriptive statistics. A sensitivity analysis excluding REMOVE CANDIDATE studies should be reported.

The authors should provide a quality-assessment rubric, preferably as a table, with explicit criteria and scoring logic. They should also provide an item-level supplementary matrix listing each study’s primary dimension, secondary flags, quality class, epistemic segment, item type, year, venue/preprint status, and key extracted findings.

5. The synthesis strategy needs stronger methodological grounding.

The manuscript states that meta-analysis was not feasible due to heterogeneity and that a structured synthesis combining frequency distributions and qualitative synthesis was adopted. This is reasonable. However, the manuscript does not describe the analytical logic of the synthesis in enough detail. Terms such as “strong consensus,” “moderate consensus,” “converges,” “systematically,” and “reliably” appear throughout the Results and Discussion, but the manuscript does not define the thresholds or evidence conditions required for these labels.

For example, Section 3.2 refers to strong-consensus and moderate-consensus patterns across several studies. Section 3.3 synthesizes human–AI interaction around interaction quality and trust calibration. Section 4 claims that the socio-technical equilibrium framework was tested against rival theoretical readings. These are important analytical moves, but they are currently described narratively rather than procedurally.

The authors should add a subsection explaining how patterns were identified, how contradictory findings were handled, how “strong” versus “moderate” consensus was determined, how dimensions were coded, and how the socio-technical framework was inductively derived. If the framework was developed after coding, the authors should describe the coding-to-construct pathway. If it was used as an organizing lens during synthesis, that should be acknowledged. The claim that the framework was “tested” against rival readings also requires more methodological detail: What were the rival readings? How were they selected? What would have counted as disconfirming evidence?

6. Several central claims are overstated relative to the evidence base.

The abstract and discussion repeatedly state that AI-assisted programming “reliably accelerates” code generation but does not reduce total development effort. The first part is plausible for bounded generation tasks, but the second part is more difficult to substantiate because the manuscript itself acknowledges that few productivity studies measure speed, correctness, rework, and downstream integration in the same design. Section 3.4 explicitly states that the corpus does not support a universal acceleration claim and that evidence is stronger for conditional acceleration than for total development productivity. The abstract should reflect this more cautiously.

The manuscript also states that prompting and validation literacy emerge as “primary mediators” of performance variability. This is theoretically plausible, but “mediator” has a specific methodological meaning. Unless the corpus includes formal mediation analyses or the authors conduct a structured causal synthesis showing mediation, this should be rephrased as “important moderators,” “conditioning factors,” or “mechanisms suggested by the evidence.”

Similarly, claims about organizational impact should be moderated. Section 3.7 is based on only seven studies, and Section 3.8 is based on one primary economic-impact study. Yet the discussion sometimes treats organizational governance and economic implications as if they are well-supported system-level findings. The manuscript does acknowledge the thinness of these dimensions, but the interpretive language should remain consistently cautious.

7. The Results section sometimes blends findings, interpretation, and framework-building.

The Results section is detailed and informative, but it often moves beyond reporting synthesized evidence into theoretical interpretation. For example, Section 3.2.7 concludes that the code-quality evidence provides grounding for the verification-overhead component of the socio-technical equilibrium framework. Section 3.3.8 similarly positions interaction quality and trust calibration as primary constructs. Section 3.8 describes Economic Impact as the aggregate expression of the whole socio-technical equilibrium.

These claims may be useful, but the current structure risks making the Results section an extension of the Discussion. The authors should decide whether Section 3 is primarily an evidence synthesis by dimension or a construct-development section. A cleaner structure would be: first, present dimension-level findings with careful evidence grading; second, introduce the framework in a separate synthesis section; third, discuss theoretical and practical implications. This would make the argument easier to follow and reduce repetition.

8. Figures and tables need improvement for readability and analytical value.

Figure 1, the PRISMA dual-pathway flow diagram, is visually dense and difficult to read at manuscript scale. It contains important information, but much of it is too small to function effectively in the paper. The authors should simplify the figure, enlarge the text, and move detailed pathway notes to an appendix.

Figures 2 and 3 use donut charts. These are visually attractive but not the most data-efficient way to present categorical distributions. A bar chart or table may be clearer, especially for readers comparing dimensions and quality classes. Figure 4 is useful but needs clearer labeling and a caption that explains whether percentages are calculated within each analytical dimension. Figure 5 should make the partial-year nature of 2026 visually explicit, not only in the caption. Figure 7 is potentially important, but the manuscript needs to explain how binary dimension flags were assigned and why the diagonal reflects “flagged in any role” rather than self-correlation. Figure 8 is central to the theoretical contribution, but it appears relatively late in the Discussion; the framework should either be introduced earlier or previewed more clearly before Results. Figure 9 is useful for transparency, but it would be more informative if linked to quality class and epistemic segment.

Table 1 should be revised, as noted above, to distinguish structured-search records from final post-supplementary totals. The paper would also benefit from additional tables: one summarizing the seven dimensions and their evidence strength, one defining the five socio-technical equilibrium constructs, and one mapping major claims to the types of evidence supporting them.

9. The discussion is intellectually strong but too expansive and repetitive.

The Discussion is the most ambitious part of the manuscript, but it is also where the manuscript becomes rhetorically heavy. The central argument—that AI-assisted programming redistributes work rather than simply removing it—is repeated many times across Sections 4.1 to 4.10. The repeated contrast between generation efficiency and verification overhead is useful, but it should be streamlined.

The current Discussion would be stronger if it were shortened and reorganized around fewer, sharper contributions. The authors could combine Sections 4.1 and 4.3, merge parts of Sections 4.4 and 4.6, and reduce the length of Section 4.10. The “rival readings” section is promising, but it should be more concise and more explicitly tied to the methodological synthesis. The manuscript should avoid giving the impression that the framework has been validated more strongly than the review design permits.

10. Ethics, conflicts of interest, and transparency statements require careful editing.

The Conflict of Interest section contains several serious editorial problems. The sentence “My sole pourpose is to stablsh the actual state of the impact of the AI-assisted...” is repeated, contains typographical errors, and uses inappropriate wording such as “over-sought.” The section should be rewritten in standard academic form: “The author declares no competing interests.” If the author has no financial, professional, or personal relationship with companies whose tools are reviewed, this can be stated briefly and correctly.

The ethics and transparency discussion should also be expanded. Although this is a literature review and does not involve direct human-subject data collection, the manuscript uses AI-assisted tools in the review process. The authors should clearly disclose AI use, data-handling procedures, availability of supplementary artifacts, and whether the review protocol and extraction sheets are deposited in a stable repository. If

proprietary AI tools were used to process copyrighted articles, the authors should clarify whether full texts were uploaded to external systems and how data privacy/copyright risks were managed.

Minor Comments

The title is precise and appropriate, but it may be slightly broad. If the paper's distinctive contribution is the socio-technical equilibrium framework, the current subtitle is useful and should be retained.

The abstract is informative but too long and too assertive in places. It should state more clearly that this is a systematic literature review with structured qualitative synthesis, not a meta-analysis. It should also clarify that economic evidence is extremely limited and that several conclusions are conditional.

The keyword list is appropriate, although "socio-technical equilibrium" may be too manuscript-specific unless the authors expect it to become a named construct. Consider adding "large language models," "code generation," or "AI coding assistants."

The manuscript does not include highlights. If the target journal requires them, they should be added; if not, this is not a problem.

The phrase "AI Assisted Programming in SE" in the running header should be standardized as "AI-Assisted Programming in Software Engineering."

Several typographical and stylistic errors require correction: "pourpose," "stablish," "over-sought," and the repeated sentence in the Conflict of Interest section. The notation "ChatGPT (4o/o1 family)" should be clarified and formatted correctly.

The manuscript uses British spelling in several places, such as "synthesised" and "organisational." This is acceptable if applied consistently.

The references include a high proportion of arXiv preprints. This is understandable in a fast-moving field, but the manuscript should ensure that preprint status is always transparent and that journal/conference versions are cited where available.

Claims Requiring Citation, Qualification, or Removal

The statement in the Introduction that AI-assisted programming has moved from "experimental tooling to standard development infrastructure" should be qualified. It may be true in some firms and communities, but adoption is uneven across organizations, domains, countries, regulatory environments, and project types.

The abstract claim that "no integrative account has explained why gains in one dimension so consistently coincide with costs in another" is too sweeping. The authors can say that prior reviews have not fully integrated technical, human, organizational, and economic evidence into a unified explanatory framework, but they should avoid implying that no relevant socio-technical explanation exists.

The claim that productivity gains “diminish or reverse as complexity and contextual dependencies increase” is plausible but should be phrased as a tendency observed across certain study designs rather than as a universal pattern.

The phrase “prompting and validation literacy emerge as primary mediators of performance variability” should be revised unless mediation evidence is formally reported. “Conditioning factors” or “mechanisms” would be more defensible.

The claim that “multi-agent configurations, despite higher autonomy, do not outperform single-agent approaches on most evaluated tasks” should specify the studies, task types, and benchmarks. It should not be generalized to all multi-agent configurations.

The conclusion’s claim that the equity dimension of cognitive redistribution tracks racial, gender, and accessibility lines should be moderated. The evidence appears early and limited. The claim is important, but the manuscript should avoid presenting it as a settled corpus-level conclusion.

The claim that organizations that scale AI assistance successfully are those that build repositories, review routines, and competence models around it is plausible but rests on limited organizational evidence. It should be presented as an implication or hypothesis rather than a demonstrated conclusion.

Redundancies, Repetitions, and Terminology Issues

The manuscript repeatedly states that AI-assisted programming redistributes work rather than eliminating it. This is the manuscript’s central insight, but the repetition weakens the prose. The authors should state the idea clearly once in the Introduction, develop it through the Results, and return to it in the Discussion without restating it in nearly identical form across multiple sections.

The terms “generation efficiency,” “productivity,” “candidate code,” “accepted output,” “validated output,” and “net development productivity” should be defined more sharply. At times, the manuscript moves between these concepts too quickly.

The terms “verification overhead,” “validation cost,” “review burden,” “checking,” and “remediation” are related but not identical. The framework would benefit from a short construct-definition table.

The terms “AI-assisted programming,” “AI-assisted coding,” “LLM-based coding assistants,” “AI coding assistants,” “agentic systems,” and “multi-agent configurations” should be used more consistently. If these refer to different classes of tools, the distinctions should be explicit.

The labels “strong consensus” and “moderate consensus” should not be used unless formally defined.

The phrase “tested against rival theoretical readings” is rhetorically strong but methodologically ambiguous. The authors should either explain the testing procedure or soften the language to “examined in relation to alternative interpretations.”

Final Recommendation to the Authors

This is a promising and potentially valuable manuscript, but it requires major revision before it can be considered publishable. The topic is important, the corpus is ambitious, and the socio-technical equilibrium framework has real potential. However, the review methodology must be made more transparent and internally consistent, especially regarding AI-assisted extraction and synthesis. The quality-assessment framework must be operationalized, record counts must be clarified, claims must be calibrated to the strength of evidence, and the Discussion should be tightened substantially.

If the authors provide a reproducible review protocol, reconcile the AI-use contradiction, define the evidence-classification scheme, add item-level supplementary materials, and moderate the strongest claims, the manuscript could become a strong contribution to empirical software engineering research.

Confidential Comments to the Editor

The manuscript is timely and potentially suitable for *Empirical Software Engineering*, but I recommend major revisions. The main risk is methodological credibility. The authors claim that AI tools were not used for extraction or synthesis, yet the threats-to-validity section states that NotebookLM, ChatGPT, and Elicit supported thematic exploration and extraction-related tasks. This contradiction must be resolved before the paper can be evaluated fairly. The review also relies on under-specified quality classifications and an opaque AI-assisted search pipeline. I do not see these problems as fatal if the authors can provide a complete audit trail, extraction matrix, and revised methods section. The conceptual contribution is promising, but the evidentiary basis and claim calibration need substantial strengthening.